

SUNNANDÆG

ONE SINE, THREE FOLDS

USER GUIDE · VERSION 1.0 · MAC

K.ODK

One sine, three folds

Sunnandæg is one sine oscillator through three **folding** stages, fed back into themselves, then **Twin Peak** (two resonant peaks) or a plain low-pass, the **Rez EQ** and the output. Four pads move it; a sample & hold, a shift register and two slopes move the rest.

It comes three ways: **ORIGIN** (the circuit, played on the pads), **SYNTH** (eight voices, played from a keyboard) and **MODULAR** (every inner connection cut: you patch it with cables).

Too much to read? Skip it. Open SETTING, hold RND+FD in Presets for a moment and let go. Drag the four pads. Hold a slot to keep what you hear.

In your DAW

The app, the AU (AUv3) and the VST3 are the same instrument. In your DAW you'll find **Sunnandæg** under *k.odk*. In ORIGIN it plays on its own; MIDI notes set its pitch. SYNTH plays from MIDI notes.

Installing

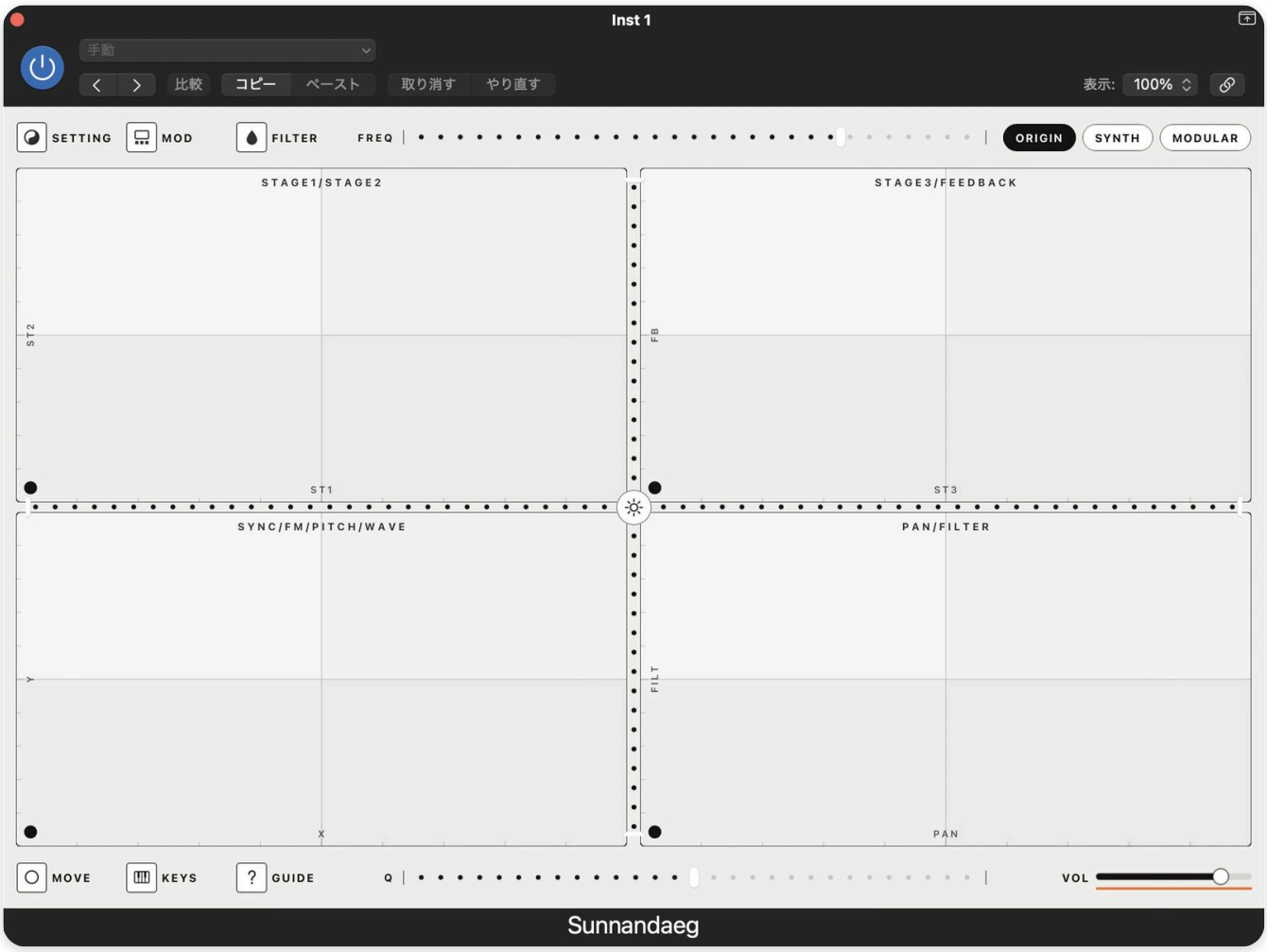
1. Open **Sunnandæg.pkg** and follow the installer. You can choose: **Standalone + AU**, and/or **VST3**.
2. Open **Sunnandæg** once from Applications — this registers the AUv3.
3. In your DAW, rescan plug-ins. Sunnandæg is under **k.odk**.

The camera is optional: the first time the sun is turned on, macOS asks for permission.

Standalone + AU	The Sunnandæg app in /Applications, with its Audio Unit (AUv3) inside: Logic, GarageBand, Live and other AU hosts list it under *k.odk*.
VST3	In /Library/Audio/Plug-Ins/VST3 — Live, Bitwig, Reaper, Cubase … The VST3 has inputs: CV IN (the Setting panel).
The app	Plays on its own, through the Mac's output. If that output goes away (headphones unplugged, a Bluetooth device taken by another device), it opens its output again by itself.

Requires macOS 12 or later · Apple silicon and Intel.

The two bars



SETTING · MOD · FILTER	The panels: the engine, the output, the camera, CV in and the presets · the modulators · the filters (ORIGIN only). **Done** or Esc closes them.
ORIGIN · SYNTH · MODULAR	The mode. Each mode keeps its own sound and its own presets.
FREQ · Q	The long dotted sliders at the top and the bottom: the oscillator's frequency and the filter's Q. In SYNTH: CUT and RES.
MOVE	The four pads recorded together (12 s). Tap: record, loop, stop. Double tap: clear.
KEYS	The keyboard along the bottom (always out in SYNTH). **Z −** and **+ X** move it an octave; the computer's keys A W S E D F T G Y H U J K play it. Lower on a key: louder.
VOL	The output volume, with the level under it. Double-click: 80 %.

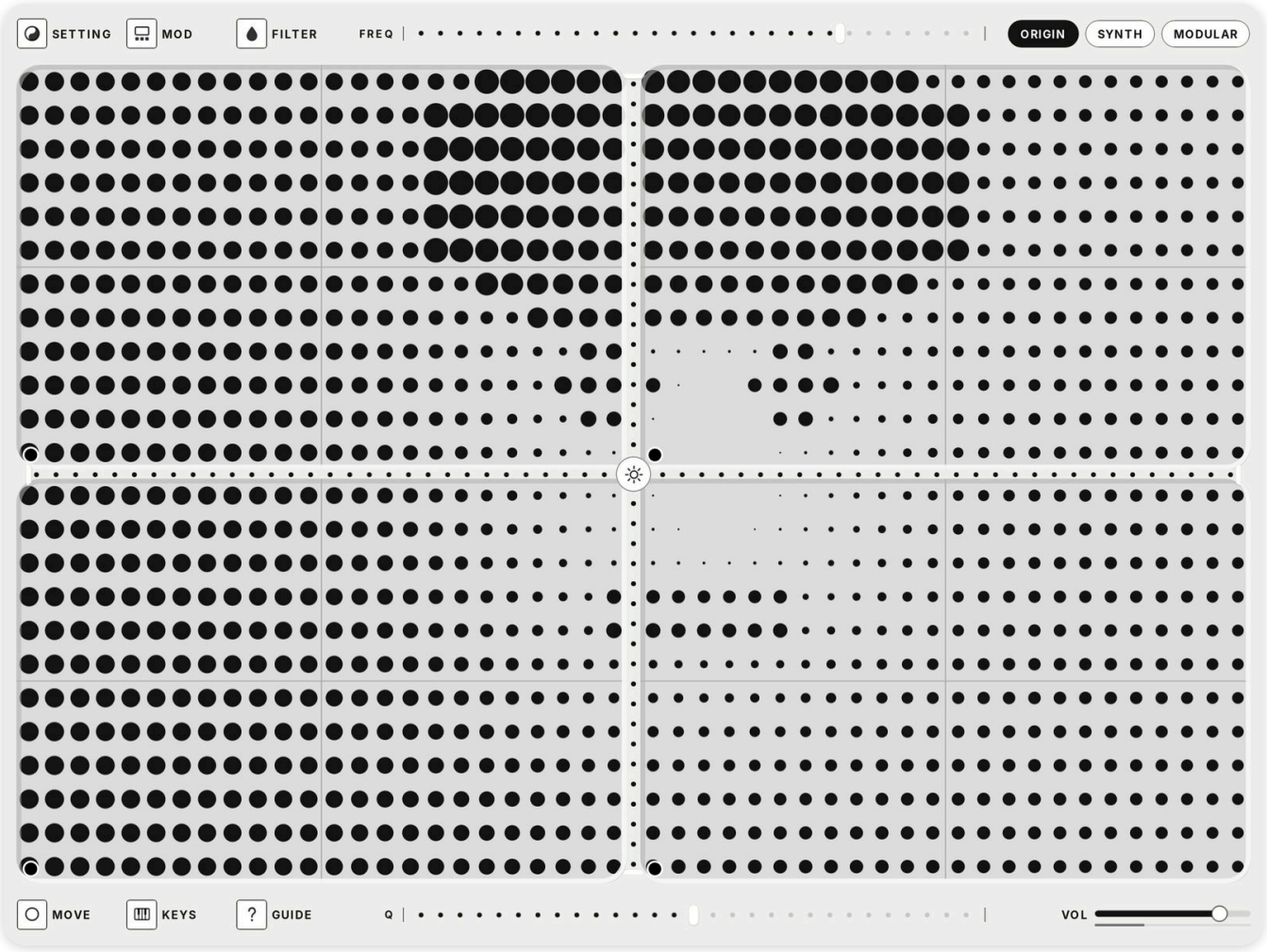
Four pads, a cross, the sun

Each pad moves two of the circuit's knobs: its name on top, X along the bottom, Y up the left side.

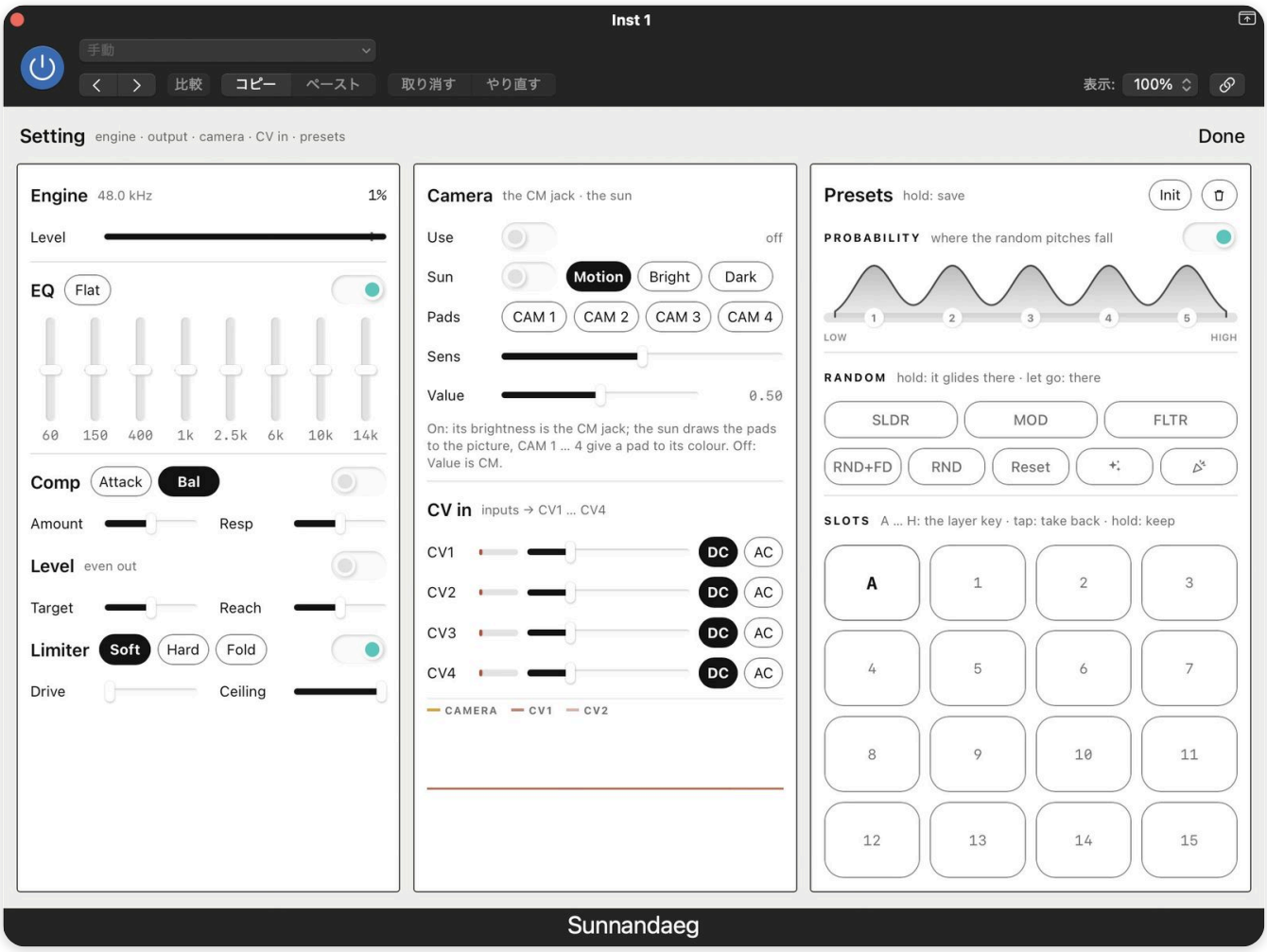
STAGE1/STAGE2	The first two folds: ST1 (X) and ST2 (Y).
STAGE3/FEEDBACK	The third fold, ST3 (X), and the feedback, FB (Y).
SYNC/FM/PITCH/WAVE	The feedback's mode: where it goes back into the oscillator.
PAN/FILTER	PAN (X) and FILT (Y): the filter.
The cross	The dotted sliders between the pads: left, the shift register's speed · right, the S&H's divide · up, the S&H's threshold · down, the shift register's threshold. They are dark while the S&H and the shift register are on.

The sun and the camera

The round key where the pads meet. Tap it: the camera's picture takes the pads, and each pointer is drawn to the cell that moves most (SETTING → Camera: or the brightest / the darkest). The pads take no hand meanwhile. Tap again to stop.

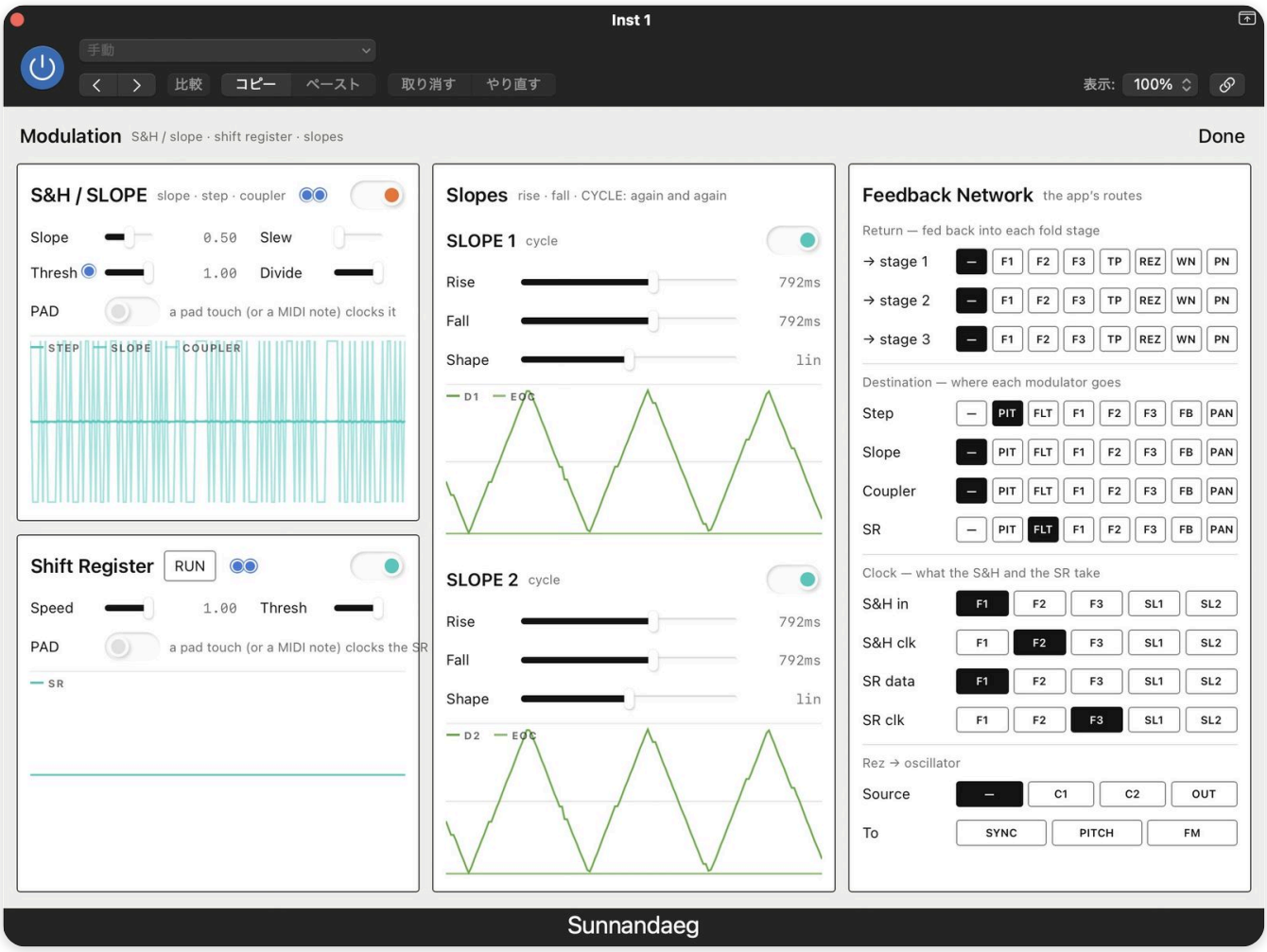


Setting



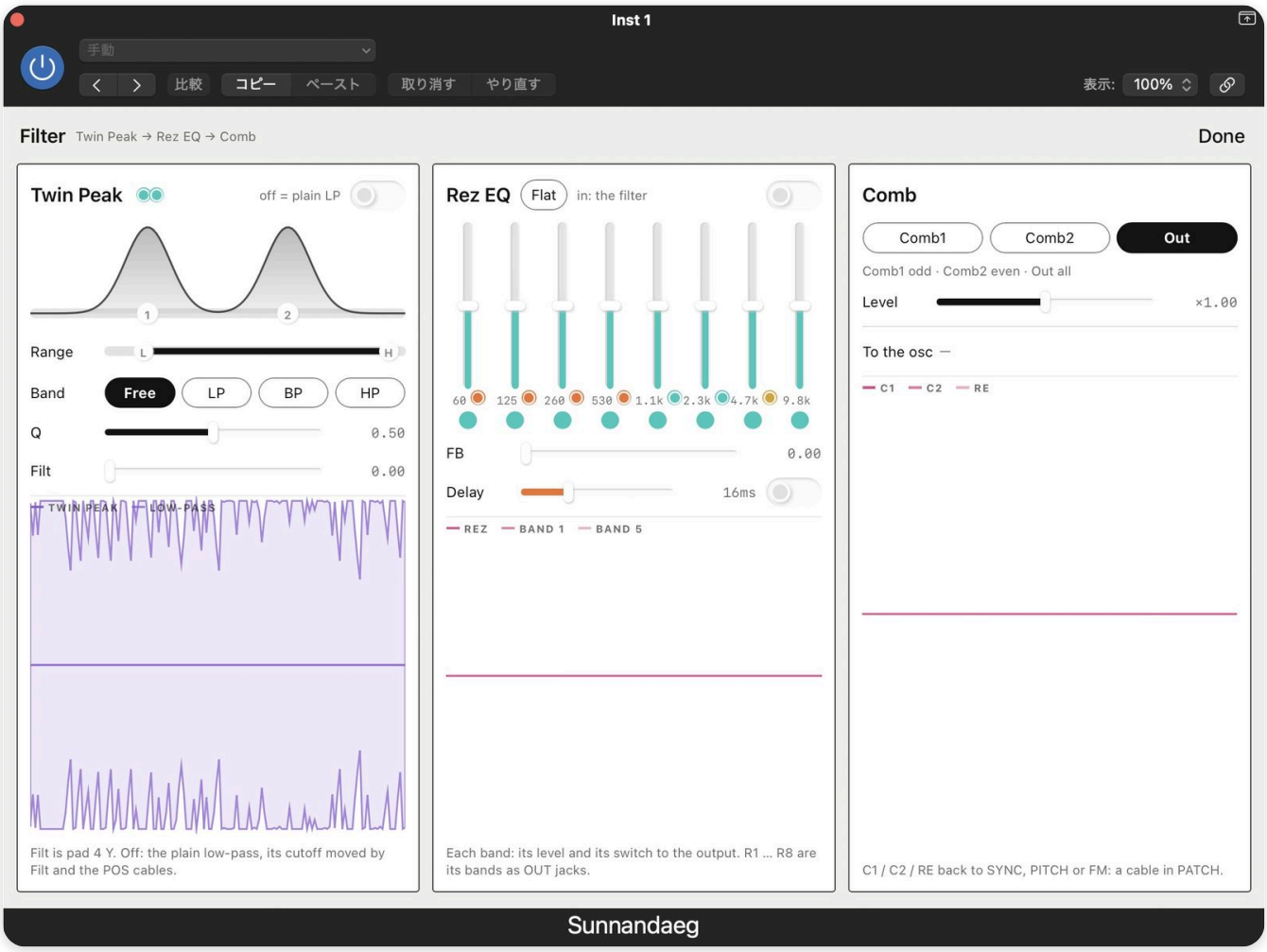
Engine	The sample rate, the load and the Level meter. **EQ** (eight bands, Flat), **Comp** (Attack / Bal, Amount, Resp), **Level** (even out: Target, Reach), **Limiter** (Soft / Hard / Fold, Drive, Ceiling).
Camera	Use: its brightness is the CM jack (off: Value is CM). Sun: the picture draws the pads — Motion, Bright or Dark. Pads: CAM 1 … 4 give one pad to the picture's colour. Sens.
CV in	VST3: its inputs as CV1 … CV4, a gain each, DC (as they come) or AC.
Presets	PROBABILITY, RANDOM and the SLOTS — see Presets and random.

Modulation



S&H / SLOPE	Slope, Slew, Thresh, Divide. PAD: a pad touch (or a MIDI note) clocks it. Its scope: step, slope, coupler.
Shift Register	RUN, Speed, Thresh, PAD.
Slopes	SLOPE 1 and 2: Rise, Fall, Shape, and cycle (again and again).
Feedback Network	ORIGIN: the routes on chips — Return into each fold stage, Destination of each modulator, Clock of the S&H and the shift register, and the Rez back to the oscillator. In SYNTH and MODULAR these are cables.

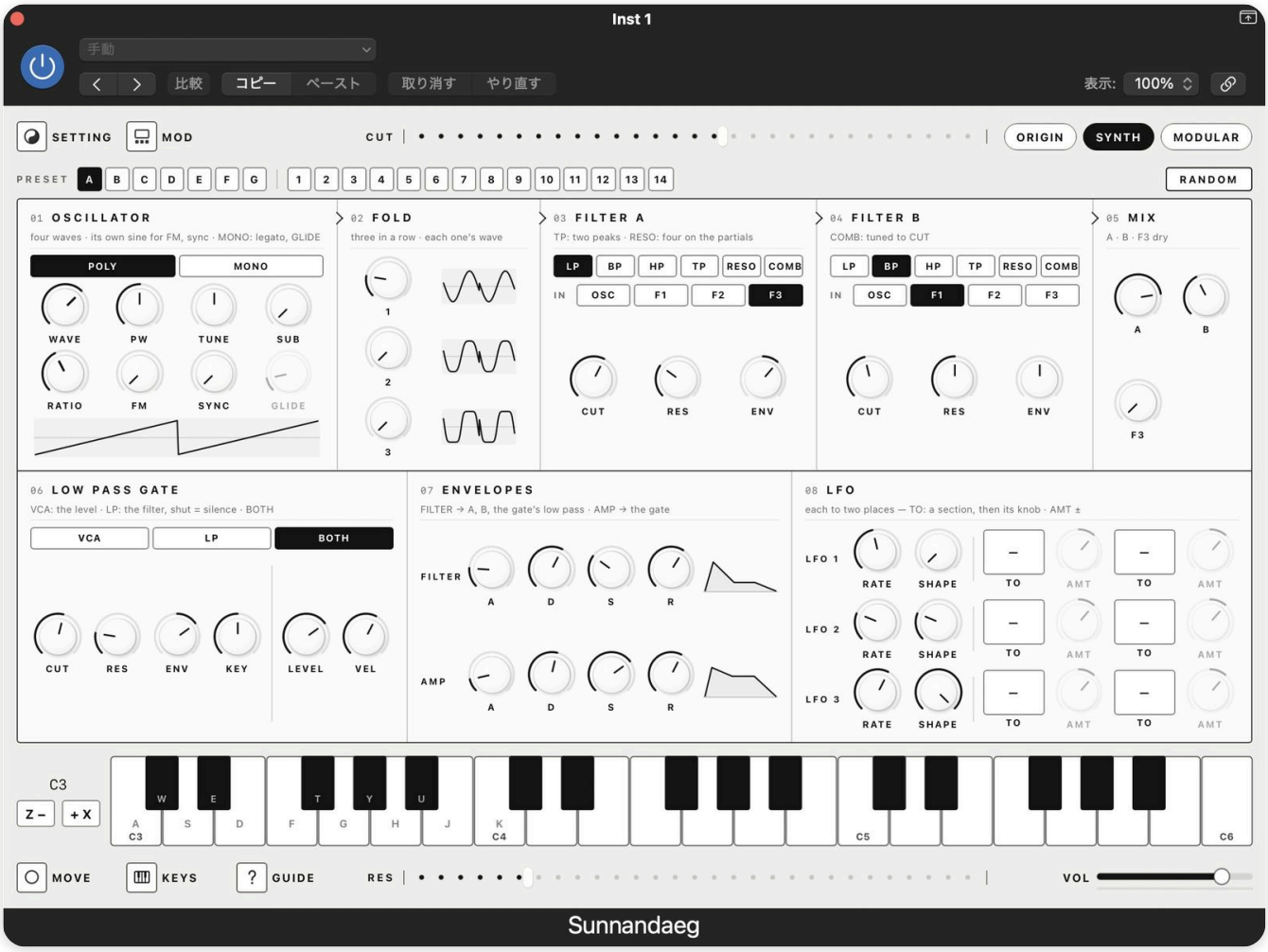
Filter



Twin Peak	Two peaks (drag 1 and 2), Range (L … H), Band (Free · LP · BP · HP), Q, Filt. Off: the plain low-pass.
Rez EQ	Eight bands, each a level and a switch to the output. Flat, FB, Delay.
Comb	Comb1 (the odd bands) · Comb2 (the even) · Out (all), and its Level.

Eight voices

The sections in the order the sound goes — the arrows between them show the way. Touch a knob: its name and value in its section's head. Double-click: its first value. The preset row above the face (A … G, 1 … 14) is SYNTH's own: tap to take back, hold to keep, ⌥-click to clear.



How it is built

Each voice is ORIGIN's circuit as a playable synthesizer: the same oscillator and three folds, two filters instead of Twin Peak and the low-pass, and a low pass gate at the end — played from keys.

OSCILLATOR → FOLD 1 · 2 · 3 → FILTER A · FILTER B → MIX → LOW PASS GATE → out

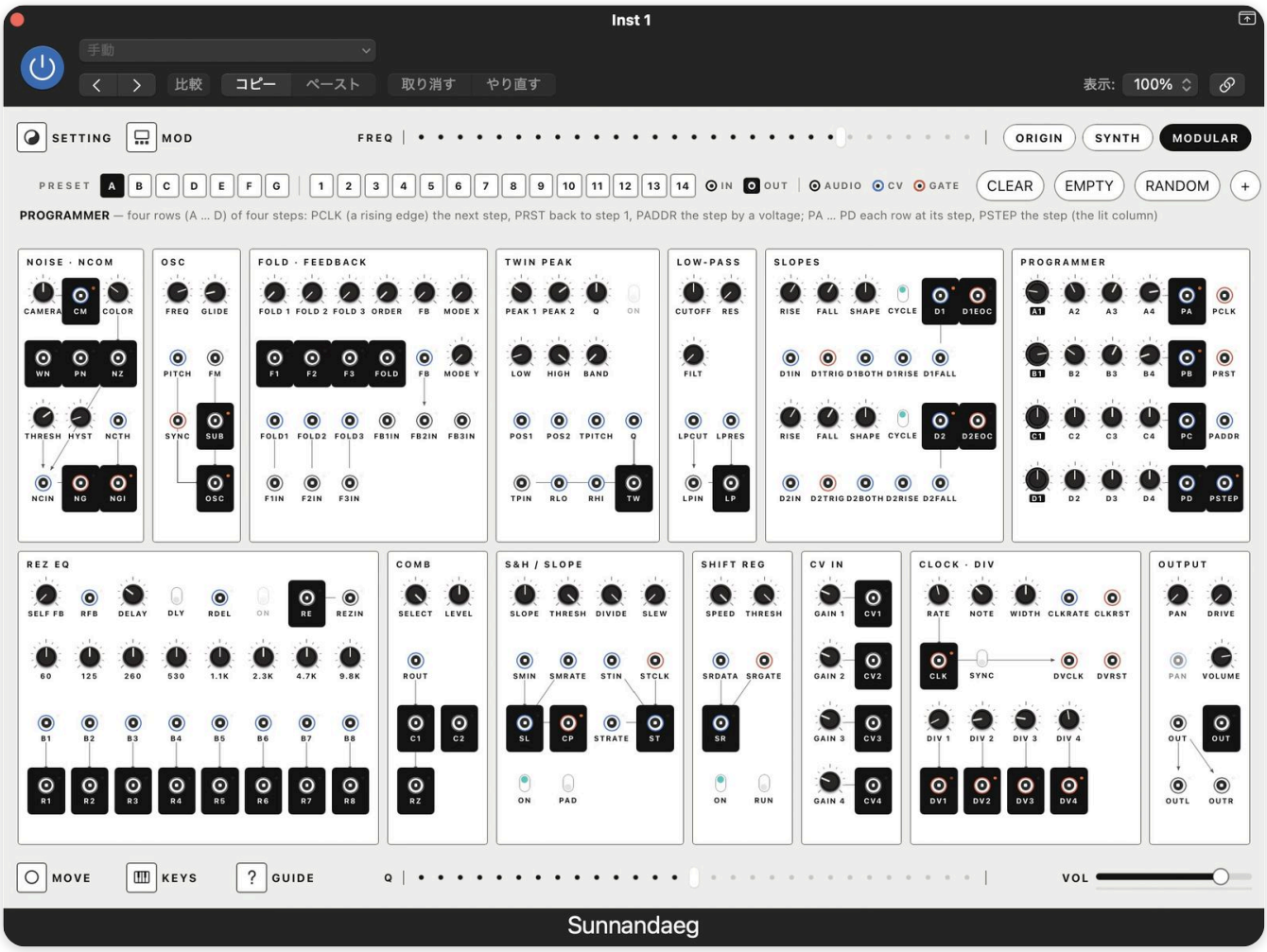
Each filter chooses its input — the oscillator (OSC) or a fold stage (F1, F2, F3) — and its kind. MIX sets A, B and the third fold dry; the gate then shapes the level, its low pass, or both.

01 · OSCILLATOR	Four waves, its own sine for FM, and sync. **POLY** (eight voices) or **MONO** (legato, with GLIDE). WAVE, PW, TUNE, SUB · RATIO, FM, SYNC, GLIDE. Its wave is drawn beneath.
02 · FOLD	Three folds in a row: 1, 2, 3, each with its wave.
03 · FILTER A	LP · BP · HP · TP (two peaks) · RESO (four, on the partials) · COMB. IN: OSC, F1, F2 or F3. CUT, RES, ENV.
04 · FILTER B	The same; COMB is tuned to CUT.
05 · MIX	A, B and F3 dry.
06 · LOW PASS GATE	VCA (the level) · LP (the filter: shut is silence) · BOTH. CUT, RES, ENV, KEY · LEVEL, VEL.
07 · ENVELOPES	FILTER (to A, B and the gate's low pass) and AMP (to the gate), each A · D · S · R.
08 · LFO	Three, each RATE and SHAPE and two places: **TO** opens a list (a section, then its knob), AMT sets how much, ±.
RANDOM	A new sound: pluck, pad, bass, lead, drone or bell, and around it.

What moves what

FILTER env	Filters A and B and the gate's low pass, each by its own ENV (±).
AMP env	The low pass gate. VEL: how much a key's velocity counts. KEY: the gate's cutoff follows the note.
LFO 1 · 2 · 3	Each moves two knobs of the face, chosen with TO; AMT ± sets how much.
POLY	Eight voices, one per key.
MONO	One voice, legato, gliding to each new key by GLIDE.

Only the cables



Every inner connection is cut. **OUT** jacks sit on black plates, **IN** jacks on the white; the ring's colour says what it carries — **AUDIO**, **CV** or **GATE**.

How the case is laid out

The top row is the sound path and its sources: NOISE · NCOM, OSC, FOLD · FEEDBACK, TWIN PEAK, LOW-PASS, SLOPES and PROGRAMMER. The bottom row: REZ EQ and COMB, the S&H / SLOPE, the SHIFT REG, CV IN, CLOCK · DIV and OUTPUT. Any OUT may go into any IN. A cable's middle sets its depth (±). **RANDOM** builds a new patch over the case; **+** adds two more cables to it.

A first patch

1	OSC **OSC** → FOLD · FEEDBACK **F1IN**, FOLD · FEEDBACK **FOLD** → LOW-PASS **LPIN**.
2	LOW-PASS **LP** → OUTPUT **OUTL** and **OUTR**. Now it sounds; turn up FOLD 1.
3	SLOPES: CYCLE on for the first, then **D1** → LOW-PASS **LPCUT**: the filter sweeps by itself.

Cables	Drag from an OUT to an IN. An IN takes many cables (they add); an OUT feeds many. Drag a cable's middle up or down: how much (±). Pull a plug off its IN, or double-click it: gone. A knob sets where an IN starts; its cables move it from there.
CLEAR EMPTY RANDOM +	**CLEAR** and **EMPTY**: every cable off. **RANDOM**: a new patch. **+**: two more cables onto the patch there is.
The line	Under the bar: what the module or the jack under the pointer does.

The modules

NOISE · NCOM	CAMERA (the CM jack), COLOR, THRESH, HYST. OUT: WN, PN, NZ, NG, NGI. IN: NCIN, NCTH.
OSC	FREQ, GLIDE. IN: PITCH, FM, SYNC. OUT: SUB, OSC.
FOLD · FEEDBACK	FOLD 1 … 3, ORDER, FB, MODE X, MODE Y. OUT: F1, F2, F3, FOLD. IN: FB, FOLD1 … 3, FB1IN … FB3IN, F1IN … F3IN (each stage's input).
TWIN PEAK	PEAK 1, PEAK 2, Q, ON, LOW, HIGH, BAND. IN: POS1, POS2, TPITCH, Q, TPIN, RLO, RHI. OUT: TW.
LOW-PASS	CUTOFF, RES, FILT. IN: LPCUT, LPRES, LPIN. OUT: LP.
SLOPES	Two: RISE, FALL, SHAPE, CYCLE. OUT: D1, D1EOC, D2, D2EOC. IN: IN, TRIG, BOTH, RISE, FALL.
PROGRAMMER	Four rows of four (A1 … D4). OUT: PA … PD, PSTEP. IN: PCLK, PRST, PADDR.
REZ EQ	SELF FB, DELAY, DLY, ON, and eight bands 60 … 9.8K. IN: RFB, RDEL, REZIN, B1 … B8. OUT: RE, R1 … R8.
COMB	SELECT, LEVEL. IN: ROUT. OUT: C1, C2, RZ.
S&H / SLOPE	SLOPE, THRESH, DIVIDE, SLEW, ON, PAD. IN: SMIN, SMRATE, STIN, STCLK, STRATE. OUT: SL, CP, ST.
SHIFT REG	SPEED, THRESH, ON, RUN. IN: SRDATA, SRGATE. OUT: SR.
CV IN	GAIN 1 … 4. OUT: CV1 … CV4.
CLOCK · DIV	RATE, NOTE, WIDTH, SYNC, DIV 1 … 4. IN: CLKRATE, CLKRST, DVCLK, DVRST. OUT: CLK, DV1 … DV4.
OUTPUT	PAN, DRIVE, VOLUME. IN: PAN, OUT, OUTL, OUTR. OUT: OUT.

Hover a module or a jack: what it does. The dot by an OUT shows what it does now.

Keep it, draw it anew

SLOTS	In Setting: eight layers (A … H, the first key: tap for the next) of fifteen. Hold a slot: keep what you hear. Tap: take it back. The bin, then a slot: clear it. A preset keeps the knobs, the cables and MOVE.
Init	Every knob and the cables to their first state.
SLDR MOD FLTR	RANDOM, held: that section drawn anew. Switches change at once, values glide there while held; let go: there.
RND+FD RND Reset	All of the playing drawn anew · a little, and the routes anew · the effects and the returns stripped.
✦ · NEXT	✦: a nudge, here and there. NEXT: a move while held.
PROBABILITY	Where the random pitches fall: five points over LOW … HIGH, each a hump. Drag a point. Its switch off: even.
A … G · 1 … 14	SYNTH and MODULAR: their own presets in the row above the face. Tap: take back · hold: keep · ⌥-click: clear.

Tips

Folds	Raise ST1, ST2 and ST3 a little at a time; FB feeds the end back — try the four feedback modes on pad 3.
The cross	Turn on the S&H and the shift register (MOD) and drag the cross: the pads start moving by themselves.
A patch	MODULAR: RANDOM, then + a few times; drag a cable’s middle to set how much.
In a DAW	Automate the pads (Pad 1 X … Pad 4 Y) and Volume; the clock follows the DAW with SYNC on.

Credits

Sunnandæg — k.odk (Kei Odaka) · instagram.com/k_odk_

Requires macOS 12 or later · Apple silicon and Intel. Questions: instagram.com/k_odk_ · kodk.gumroad.com